

class CourseRepositoryTest:

The screenshot displays an IDE interface with the following components:

- Project Explorer (Left):** Shows a project structure with folders like `test`, `ru.natali.courses`, `config`, `controller`, `it`, `repository`, `service`, `resources`, and `screens`. The `CourseRepositoryTest` file is highlighted under the `repository` folder.
- Code Editor (Center):** Displays the `CourseRepositoryTest.java` file. The code includes annotations like `@DataJpaTest`, `@AutoConfigureTestDatabase`, `@Testcontainers`, `@ActiveProfiles("test")`, and `@Transactional`. It defines a `CourseRepositoryTest` class with methods `setUp()` and `shouldFindAllCourses()`.
- Run Console (Bottom):** Shows the execution of the `CourseRepositoryTest` class. It reports that 11 tests passed in 557 ms. The test results are listed on the left, and the log output is shown on the right.

Test Results:

- ✓ `shouldReturnEmptyListWhenNo` 434 ms
- ✓ `shouldUpdateOnlySpecificFields()` 15 ms
- ✓ `shouldFindAllActiveCourses()` 20 ms
- ✓ `shouldUpdateExistingCourse()` 12 ms
- ✓ `shouldSaveNewCourse()` 9 ms
- ✓ `shouldFindAllCourses()` 13 ms
- ✓ `shouldFindCourseById()` 13 ms
- ✓ `shouldDeleteCourse()` 9 ms
- ✓ `shouldNotFindNonExistingCourse` 10 ms
- ✓ `shouldArchiveCourse()` 10 ms
- ✓ `shouldNotFailWhenDeletingNonE` 12 ms

Log Output:

```
19:17:25,584 |-INFO in ch.qos.logback.classic.LoggerContext[default] - This is logback-classic version 1.2.3
19:17:25,585 |-INFO in ch.qos.logback.classic.util.ContextInitializer@67d18ed7 - Here is a list of candidates for the
19:17:25,585 |-INFO in ch.qos.logback.classic.util.ContextInitializer@67d18ed7 - org.springframework.boot.logging.logback
19:17:25,585 |-INFO in ch.qos.logback.classic.util.ContextInitializer@67d18ed7 - They will be invoked in the following
19:17:25,585 |-INFO in ch.qos.logback.classic.util.ContextInitializer@67d18ed7 - Constructed configuration object:
    .RootLogLevelConfigurator
19:17:25,590 |-INFO in ch.qos.logback.classic.util.ContextInitializer@67d18ed7 - org.springframework.boot.logging.logback
    milliseconds. ExecutionStatus=INVOKE_NEXT_IF_ANY
19:17:25,590 |-INFO in ch.qos.logback.classic.util.ContextInitializer@67d18ed7 - Trying to configure
19:17:25,591 |-INFO in ch.qos.logback.classic.util.ContextInitializer@67d18ed7 - Constructed configuration object:
```

class CourseRepositoryAdvancedTest:

The screenshot displays an IDE with a project explorer on the left, a code editor in the center, and a run console at the bottom.

Project Explorer: Shows the project structure for 'Courses'. The 'src/main/java/ru.natali.courses' package contains several sub-packages: 'aop', 'config', 'controller', 'dto', 'model', 'repository', 'service', and 'utils'. The 'model' package includes 'Course', 'Review', 'Role', and 'Student'. The 'repository' package contains 'CourseRepository'. The 'service' package contains 'CourseService'. The 'utils' package contains 'CoursesApplication'.

Code Editor: Shows the implementation of the `CourseRepositoryAdvancedTest` class. The class is annotated with `@DataJpaTest`, `@AutoConfigureTestDatabase(replace = AutoConfigureTestDatabase.Replace.NONE)`, `@Testcontainers`, `@ActiveProfiles("test")`, and `@Transactional`. The class contains the following methods:

```
20 @DataJpaTest
21 @AutoConfigureTestDatabase(replace = AutoConfigureTestDatabase.Replace.NONE)
22 @Testcontainers
23 @ActiveProfiles("test")
24 @Transactional
25 public class CourseRepositoryAdvancedTest {
26
27     @Autowired
28     private CourseRepository courseRepository;
29
30     @Autowired
31     private StudentRepository studentRepository;
32
33     @BeforeEach
34     public void setUp() {
35         studentRepository.deleteAll();
36         courseRepository.deleteAll();
37     }
38
39     @Test
40     public void shouldGetCourseStatistics() {
41         // Given
42         Course course1 = createAndSaveCourse(name: "Java Basics", active: true, archived: false);
43         Course course2 = createAndSaveCourse(name: "Spring Advanced", active: true, archived: false);
44
45         createAndSaveStudent(firstName: "John", lastName: "Doe", course1);
```

Run Console: Shows the execution of the tests. The output indicates that all tests passed successfully.

```
Run CourseRepositoryAdvancedTest x
✓ CourseRepositoryAdvan 709ms
✓ Tests passed: 6 of 6 tests - 709ms
C:\Users\natal\jdk\jbr-17.0.12-1\bin\java.exe ...
19:25:32,244 |-INFO in ch.qos.logback.classic.LoggerContext[default] - This is logback-classic version 1.5.18
19:25:32,245 |-INFO in ch.qos.logback.classic.util.ContextInitializer@68034211 - Here is a list of configurators discovered
a service, by rank:
19:25:32,245 |-INFO in ch.qos.logback.classic.util.ContextInitializer@68034211 - org.springframework.boot.logging.logback
.RootLogLevelConfigurator
19:25:32,245 |-INFO in ch.qos.logback.classic.util.ContextInitializer@68034211 - They will be invoked in order until
```

Проверка покрытия тестами:

Courses

Courses

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
ru.natali.courses.controller		9 %		0 %	25	32	66	77	23	30	0	6
ru.natali.courses.utils		0 %		n/a	2	2	7	7	2	2	1	1
ru.natali.courses.service		98 %		80 %	2	41	1	105	0	36	0	4
ru.natali.courses		37 %		n/a	1	2	2	3	1	2	0	1
ru.natali.courses.config		100 %		n/a	0	11	0	23	0	11	0	4
ru.natali.courses.dto		100 %		n/a	0	4	0	12	0	4	0	2
ru.natali.courses.aop		100 %		n/a	0	3	0	7	0	3	0	1
ru.natali.courses.model		100 %		n/a	0	2	0	3	0	2	0	1
Total	315 of 971	67 %	6 of 14	57 %	30	97	76	237	26	90	1	20

class CourseControllerTest:

The screenshot displays an IDE with the following components:

- Project Explorer (Left):** Shows a project structure with a `test` directory containing `java` (with sub-packages `ru.natali.courses.controller`, `it`, `repository`, `service`, and `CoursesApplicationTests`) and `resources` (with `screens`, `application-test.properties`, and `course_test_data.csv`). The `target` directory is also visible.
- Editor (Center):** Displays the `CourseControllerTest.java` file. The code includes imports for `ru.natali.courses.model.Course`, `ru.natali.courses.service.CourseService`, `java.time.LocalDate`, `java.util.Arrays`, `java.util.List`, `org.mockito.Mockito`, and `org.springframework.test.web.servlet.result.MockMvcResultHandlers`. The class is annotated with `@SpringBootTest` and `@AutoConfigureMockMvc`. It contains two `@Autowired` fields: `MockMvc mockMvc` and `ObjectMapper objectMapper`, and a `@MockBean` field: `CourseService courseService`.
- Run Window (Bottom):** Shows the execution of `CourseControllerTest` (ru.natali.courses.controller) in 193 ms. The test results are as follows:
 - ✓ `getActiveCourses_ShouldReturnAc` 140 ms
 - ✓ `getAllCourses_ShouldReturnListOfC` 11 ms
 - ✓ `getArchivedCourses_ShouldReturnA` 10 ms
 - ✓ `getActiveCourses_WhenNoActiveC` 14 ms
 - ✓ `getArchivedCourses_WhenNoArchiv` 10 ms
 - ✓ `getAllCourses_WhenNoCourses_Sho` 8 ms

class CourseControllerCrudTests:

repository

service

utils

CoursesApplication

resources

test

java

33

34

35

36

37

38

39

@MockBean

private CourseService courseService;

@Test

public void createCourse_ShouldReturnCreatedCourse() throws Exception {

// ...

Run CourseControllerCrudTests

CourseControllerCrudTests 226 ms

createCourse_ShouldReturnCreatedCourse 180 ms

deleteCourse_ShouldReturnDeletedCourse 16 ms

updateCourse_ShouldReturnUpdatedCourse 17 ms

updateCourse_WithNewCourse 13 ms

Tests passed: 4 of 4 tests - 226 ms

C:\Users\natali\.jdk\jbr-17.0.12-1\bin\java.exe ...

21:04:32,815 |-INFO in ch.qos.logback.classic.LoggerContext[default] - This is logback-classic version 1.5.18

21:04:32,817 |-INFO in ch.qos.logback.classic.util.ContextInitializer@24c4ddae - Here is a list of configurators discovered as a service, by rank:

21:04:32,817 |-INFO in ch.qos.logback.classic.util.ContextInitializer@24c4ddae - org.springframework.boot.logging.logback.RootLogLevelConfigurator

Проверка покрытия теста после написания тестов контроллеров:

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
ru.natali.courses.controller		31 %		0 %	18	32	47	79	16	30	0	6
ru.natali.courses.service		90 %		80 %	3	42	8	112	1	37	0	4
ru.natali.courses.utils		0 %		n/a	2	2	7	7	2	2	1	1
ru.natali.courses		37 %		n/a	1	2	2	3	1	2	0	1
ru.natali.courses.config		100 %		n/a	0	11	0	23	0	11	0	4
ru.natali.courses.dto		100 %		n/a	0	4	0	12	0	4	0	2
ru.natali.courses.aop		100 %		n/a	0	3	0	7	0	3	0	1
ru.natali.courses.model		100 %		n/a	0	2	0	3	0	2	0	1
Total	290 of 1 018	71 %	6 of 14	57 %	24	98	64	246	20	91	1	20